

Improve Generalization

Project: Responsible Plant Classifier

1. Your mission

Your learning target

I can compare training and validation evidence for model capacity.

I can explain overfitting without assuming every gap proves it.

I can use a declared selection rule instead of perfect training fit.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Feature Engineering. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Overfitting. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Train, Validation, Test. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Different failure causes require different repairs: simplicity, representative data, regularization, or better features.



LEARNING TARGET All spotted training leaves use black backgrounds. Propose a collection and comparison that tests the background shortcut without tuning on final test examples.

Builder name and experiment date:

Readiness for the connected explanation

Can a zero training error guarantee good validation performance?

Readiness for Distribution shift

Why does a test score need a population description?

My readiness explanation

This week's end goal

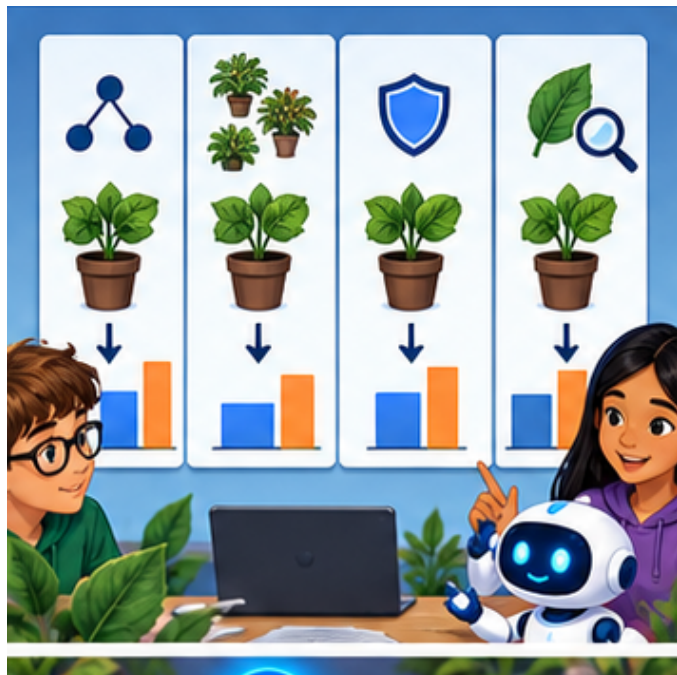
By the end, I can explain improve generalization using a traced example and a stated limitation.

2. Notice the evidence

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible evidence; do not explain more than the picture and information support.



MYSTERY The team must use improve generalization to solve a model-building problem. What information do they have? What important evidence is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what evidence would confirm or reject your prediction.

My prediction and supporting evidence:

A closer explanation

A more flexible model can represent complicated relationships and accidental details. Capacity can help when there is enough relevant evidence, but perfect training performance is not the goal. Compare training and validation using the same metric while checking leakage and representativeness.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines overfitting and model capacity.

Apply the connected idea: Distribution shift

A model trained on one set of conditions may encounter another. Lighting, measuring tools, class frequency and the relationship between inputs and labels can change. A shift does not automatically imply

failure, but it challenges the assumption that old evidence represents current use.

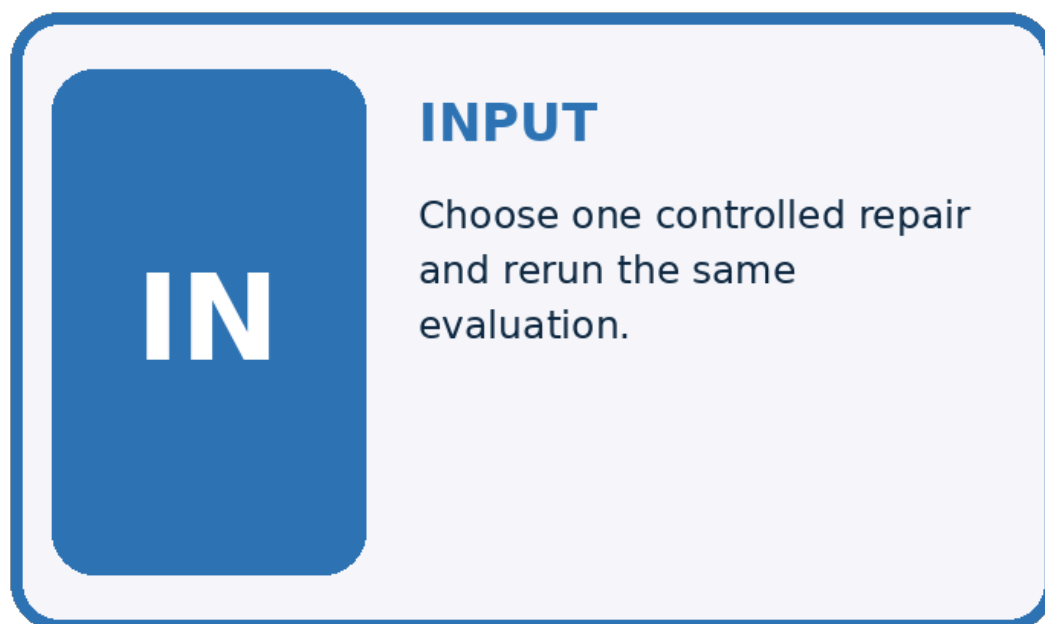
Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines distribution shift.

3. See how it works

ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.



ML

MODEL

Different failure causes require different repairs: simplicity, representative data, regularization, or better features.

OUT

EVIDENCE

Choose one controlled repair and rerun the same evaluation.

QUESTION Choose one controlled repair and rerun the same evaluation.

MODEL IDEA Different failure causes require different repairs: simplicity, representative data, regularization, or better features.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Evidence: high train score, low validation score, largest gap in low light. Hypothesis combines complexity and missing low-light coverage; choose one cause to test first.

Draw a fresh data → model → prediction → evaluation trace:

A closer explanation

On the same regression split, a constant has training MAE 5 and validation 6, a line has training 2 and validation 3, and a highly flexible curve has training 0 and validation 8. Under smallest validation MAE, choose the line. The curve has the lowest training loss but the worst validation result.

Before continuing, underline the decision rule and circle the evidence that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim. Keep the distinction between an observed result and a proposed next step.

Apply the connected idea: Distribution shift

On development-like bright images, a frozen classroom classifier makes 2 errors in 20 cases:10 percent. On a separate dim set it makes 6 errors in 20:30 percent. The pooled result is $8/40=20$ percent error. The pooled score hides a 20-percentage-point difference between conditions.

Before continuing, underline the decision rule and circle the evidence that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim. Keep the distinction between an observed result and a proposed next step.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

EVALUATION DESIGN

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL CASE

Expected data from the intended setting



BOUNDARY CASE

Value near a threshold or class boundary



MISSING CASE

Unknown field, label, or necessary evidence



SHIFTED CASE

New subgroup, device, condition, or time

NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain prediction.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change: The unchanged model predicts spotted for both red-background classes and healthy for both grey-background classes.

Prediction: It fails on spotted/grey and healthy/red in the crossed audit.

Reason: Those supplied predictions follow background rather than the intended leaf target.

5. Words that unlock the idea

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.



2

CONTROLLED CHANGE

one planned modification

3

ABLATION

remove one element to
measure its effect

4

REGRESSION TEST

protect previously passing behavior

Term · Precise meaning · My ML example

Term: generalization

Precise meaning: performance beyond training examples

My ML example: _____

Term: controlled change

Precise meaning: one planned modification

My ML example: _____

Term: ablation

Precise meaning: remove one element to measure its effect

My ML example: _____

Term: regression test

Precise meaning: protect previously passing behavior

My ML example: _____

MISCONCEPTION TO REPAIR Changing data, features, model, and threshold together is the fastest proof of improvement.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word · Meaning

Word: Capacity

Meaning: Flexibility of a model to represent patterns

Word: Overfitting

Meaning: Learning details that fail to generalize well

Word: Underfitting

Meaning: Failing to capture useful structure

Word: Generalization

Meaning: Performance on relevant unseen examples

Terms for Distribution shift

Term · Meaning

Term: Distribution shift

Meaning: A change in the data-generating conditions

Term: Monitoring

Meaning: Repeated checks of inputs, outputs and later outcomes

Term: Reference labels

Meaning: The labels used to evaluate predictions

Term: Fallback

Meaning: A planned alternative when the model is unsuitable

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

WEEK 30 • WORKED EVIDENCE

- 1 Evidence: high train score, low validation score, largest gap in low light.
- 2 Hypothesis combines complexity and missing low-light coverage; choose one cause to test first.
- 3 Add representative low-light training examples while freezing model settings.
- 4 Rerun all validation cases; compare overall, error slice, subgroup gap, and regressions.

STEP 1 Evidence: high train score, low validation score, largest gap in low light.

STEP 2 Hypothesis combines complexity and missing low-light coverage; choose one cause to test first.

STEP 3 Add representative low-light training examples while freezing model settings.

STEP 4 Rerun all validation cases; compare overall, error slice, subgroup gap, and regressions.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

A closer explanation

Overfitting and model capacity at a glance

Part · Evidence or action · Interpretation

Part: Constant

Evidence or action: Training 5

Interpretation: Validation 6

Part: Line

Evidence or action: Training 2

Interpretation: Validation 3

Part: Flexible curve

Evidence or action: Training 0

Interpretation: Validation 8

The curve result is consistent with overfitting, but a gap alone does not prove its cause. A changed population, bad labels or a split error could also produce a gap. Inspect the procedure before claiming a diagnosis. The constant may miss useful structure, but these numbers alone do not identify the true relationship.

Does a training-validation gap alone prove overfitting? Explain the steps or evidence behind your response.

Apply the connected idea: Distribution shift

Distribution shift at a glance

Part · Evidence or action · Interpretation

Part: Bright condition

Evidence or action: 2 errors of 20

Interpretation: 10 percent error

Part: Dim condition

Evidence or action: 6 errors of 20

Interpretation: 30 percent error

Part: Combined

Evidence or action: 8 errors of 40

Interpretation: 20 percent error

The comparison identifies an observed weakness, not its cause. Check how images and labels were collected and whether other factors changed with lighting. An input shift can be detected before labels arrive, but performance cannot be measured from predictions alone.

Why is 8/40 incomplete by itself? Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Improve Generalization

Test whether a plant model follows the leaf or a convenient background shortcut.

The facts we will use

Toy training photos show spotted leaves on red cards and healthy leaves on grey cards. Intended use allows both backgrounds for either class.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Improve Generalization

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Notice correlation

Background colour perfectly tracks the labels in this training sample.



2. Design crossed cases

Collect spotted/red, spotted/grey, healthy/red, healthy/grey with plant grouping preserved.



3. Inspect fixed model

Compare predictions for the same leaf on both backgrounds; do not assume an error cause before observing it.

CHANGE ONE THING

The unchanged model predicts spotted for both red-background classes and healthy for both grey-background classes.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Notice correlation

What happens and why: Background colour perfectly tracks the labels in this training sample.

Step: 2. Design crossed cases

What happens and why: Collect spotted/red, spotted/grey, healthy/red, healthy/grey with plant grouping preserved.

Step: 3. Inspect fixed model

What happens and why: Compare predictions for the same leaf on both backgrounds; do not assume an error cause before observing it.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Compare predictions for the same leaf on both backgrounds; do not assume an error cause before observing it. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

One controlled audit can reveal a shortcut in this toy setting without proving how every internal feature is represented.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



COMPARE • DIAGNOSE • REPAIR

Tempting claim: Changing data, features, model, and threshold together is the fastest proof of improvement. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: passes

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: defined behavior

Actual: _____

Likely cause / next test: _____

Evidence record: missing case

Expected: safe fallback

Actual: _____

Likely cause / next test: _____

Evidence record: shifted case

Expected: measured limit

Actual: _____

Likely cause / next test: _____

CLAIM TO CHALLENGE Changing data, features, model, and threshold together is the fastest proof of improvement.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

A simpler model is not always better. If a flexible candidate has better valid evidence under the declared rule, keep that evidence visible. Model capacity is one part of an experiment, alongside features, sample size, label quality and the split.

What extra evidence would you inspect before concluding a model memorized noise?

Apply the connected idea: Distribution shift

Do not normalize away a shift by silently changing preprocessing during evaluation. Treat a revised transformation or retrained model as a new version. Use fresh appropriate evidence for its claims and retain earlier results.

State a safe classroom response to inputs outside the supported range.

8. Practise together

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



FREEZE THE EXPERIMENT

Record question, intended use, data version, grouped split, preprocessing, features, model settings, seed, metric, predictions, errors, subgroup results, and human review before claiming improvement.

Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/preprocessing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Change one development factor at a time: simplify the candidate, improve data coverage or adjust a declared regularization setting. Refit on training and compare on validation. Do not remove difficult valid examples or repeatedly search the test for a favorable result.

Apply the connected idea: Distribution shift

For a supervised classroom prototype, define a review trigger before use, such as a new unsupported input range or a documented increase in labelled error. Pause the affected use, provide a human or non-model fallback, and investigate with new development evidence. Version any revision.

Practise with less help: Interpret a gap

Use workbook W2 for the connected example “Overfitting and model capacity”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Overfitting and model capacity. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Distribution shift

Use its named workbook W2. Identify the given inputs, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A final answer is enough

Explain the picture

What fresh evidence is needed after adding crossed training data? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Repair Planner: match four evidence patterns to targeted changes, run one change, and explain what the result does not prove.

1. Write the hypothesis and expected evidence before running.
2. Repair Planner: match four evidence patterns to targeted changes, run one change, and explain what the result does not prove.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the example.



ML BUILDER MASTERY
1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

DEFINE Use all four terms precisely: generalization, controlled change, ablation, regression test.

CALCULATE Reproduce the worked evidence with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE Changing data, features, model, and threshold together is the fastest proof of improvement.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed evidence.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

11. Connect your project

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same evidence-based capstone.



RESPONSIBLE PLANT CLASSIFIER

Improve Plant Classifier with one approved change and a full before/after evidence table.

THIS WEEK'S CHECKPOINT Improve Plant Classifier with one approved change and a full before/after evidence table.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My evidence-linked project decision:

Evidence for your project

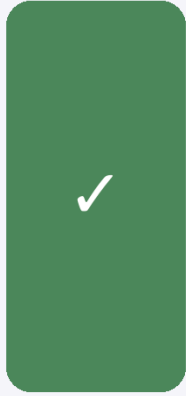
For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



INDEPENDENT ML BUILDER PROOF

Choose one controlled repair and rerun the same evaluation.

REQUIRED PROOF Choose one controlled repair and rerun the same evaluation.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent evidence and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

NEXT BRIDGE Carry the experiment evidence into: Subgroup Performance.

13. Study a complete case

Week 30 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A classifier relies on tray colour; new trays reduce performance.

The question

Propose controlled improvement.

Worked explanation

Reduce the shortcut and add representative variation; compare on fixed validation cases including new colours. Change one planned factor.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Repair a background shortcut deliberately

Evidence or stage · Value or result

Evidence or stage: Observed shortcut

Value or result: Tray colour tracks label

Evidence or stage: Changed condition

Value or result: New trays reduce performance

Evidence or stage: Revision

Value or result: Reduce shortcut and add relevant variation

Evidence or stage: Comparison

Value or result: Same validation conditions and slices

A classifier relies on tray colour; new trays reduce performance.

Propose controlled improvement.

Reduce the shortcut and add representative variation; compare on fixed validation cases including new colours. Change one planned factor.

14. Try a new case independently

Independent case: Changed independent case

Spotted leaves have blue backgrounds and healthy leaves white backgrounds in training. Design a generalisation investigation with fresh evaluation.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 30 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A team proposes more representative training examples and a simpler model. How should it compare changes while protecting the final test?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Improve Plant Classifier with one approved change and a full before/after evidence table.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

All spotted training leaves use black backgrounds. Propose a collection and comparison that tests the background shortcut without tuning on final test examples.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.